Overview

For Sprint 2, our team identified the tasks that had dependencies and organized them into a network diagram. The purpose of the diagram was to show the order of dependent work, identify the critical path, and understand which delays would have the biggest effect on the sprint schedule.

The sprint goal was to expand marketplace functionality by allowing users to filter and search listings, manage their own listings as sellers, and create or manage offers. We also included spike tasks for areas that required research before implementation.

Assumptions Used

To create the network diagram, we used these assumptions:

- user stories and spikes were labelled with story points as rough effort estimates
- integration and documentation were treated as final milestone stages
- dependencies were based on likely implementation order
- the diagram was created at the story/spike level rather than for every small checklist task

These assumptions kept the diagram readable while still showing the most important dependencies.

Tasks with Dependencies

The main sprint tasks were:

- US5: Filter Listings by Category
- US6: Search Listings by Keywords
- US9: View Own Listings as a Seller
- US10: Edit Listing
- US11: Delete Listing
- US12: Make an Offer
- US13: Cancel Pending Offer
- US16: View Sent Offers

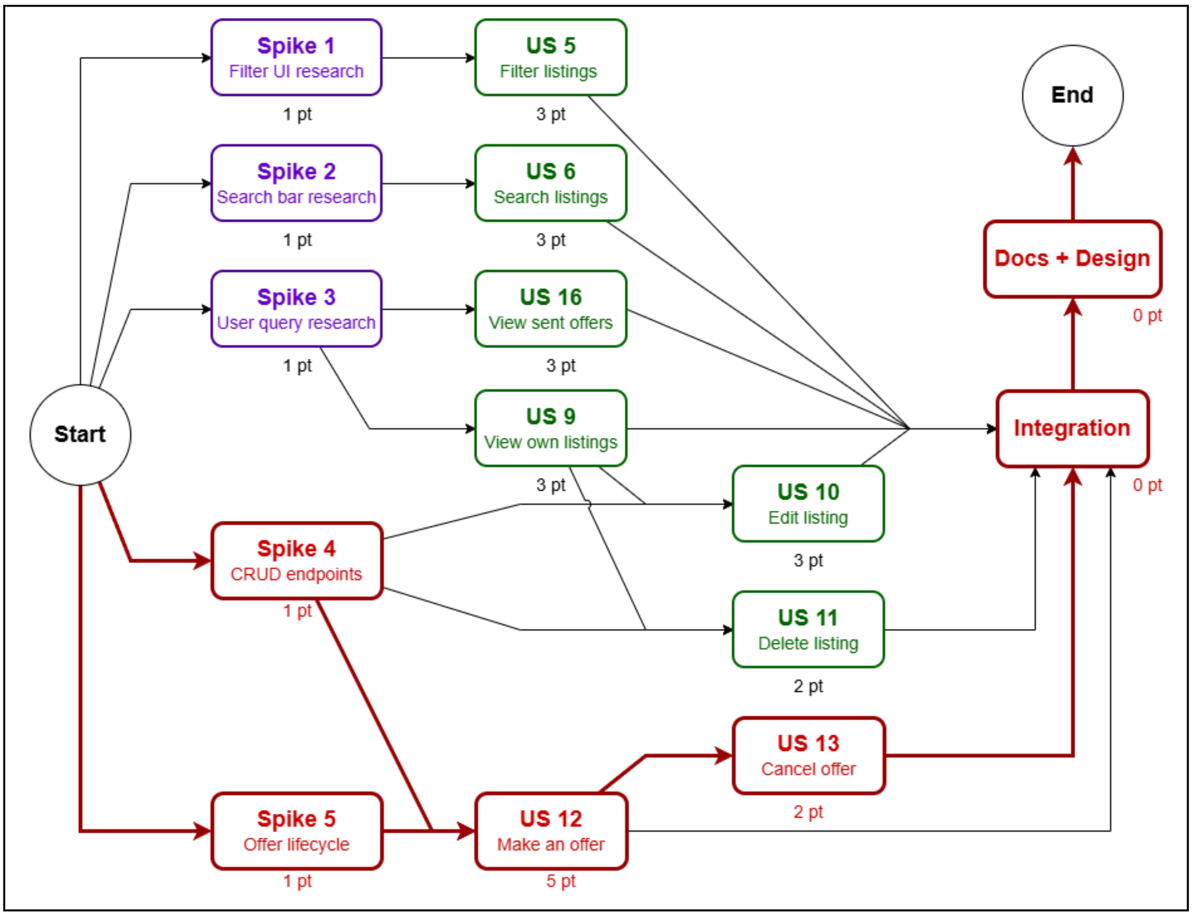
The supporting spike tasks were:

- Spike 1: Filter UI research
- Spike 2: Search bar research
- Spike 3: User query research
- Spike 4: CRUD endpoints
- Spike 5: Offer lifecycle

The main dependencies were:

- US5 depends on Spike 1
- US6 depends on Spike 2
- US9 and US16 depend on Spike 3
- US10 and US11 depend on US9 and Spike 4
- US12 depends on Spike 4 and Spike 5
- US13 depends on US12
- after implementation, the sprint moves into integration, then documentation/design, and then completion

Network Diagram



The critical path is the path shown in red

Critical Path

In our diagram, the critical path is centered on the offer workflow:

Start → Spike 4 + Spike 5 → US12 → US13 → Integration → Docs + Design → End

This branch is the critical path because it contains the longest sequence of dependent work and therefore has the greatest impact on the overall sprint schedule.

How We Plan to Stay on Schedule

To keep the sprint on schedule, we planned to complete the spike tasks early so that the more dependent stories could begin sooner. This reduced uncertainty before implementation and helped unblock the more complex stories.

We also divided work across separate branches where possible. Filtering and searching could be worked on independently from seller listing management and offer management. This allowed team members to work in parallel rather than waiting on a single sequence of tasks.

We paid special attention to the offer branch during standups because it was on the critical path. Since that part of the sprint involved new offer-related logic, it had a higher chance of creating delays. We also left time near the end of the sprint for integration and documentation so that finishing code implementation alone would not be treated as sprint completion.

Reflection

In retrospect, the offer workflow was the part of the sprint that required the most attention, which supports our decision to place it on the critical path. Working early on spikes and splitting independent stories across separate branches helped the team continue making progress. This showed that the diagram was helpful both for planning and for understanding where the main sprint risks appeared.